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Applications of AI in EEE

EEEE3134

Artificial Intelligence

Concepts and Historical Development

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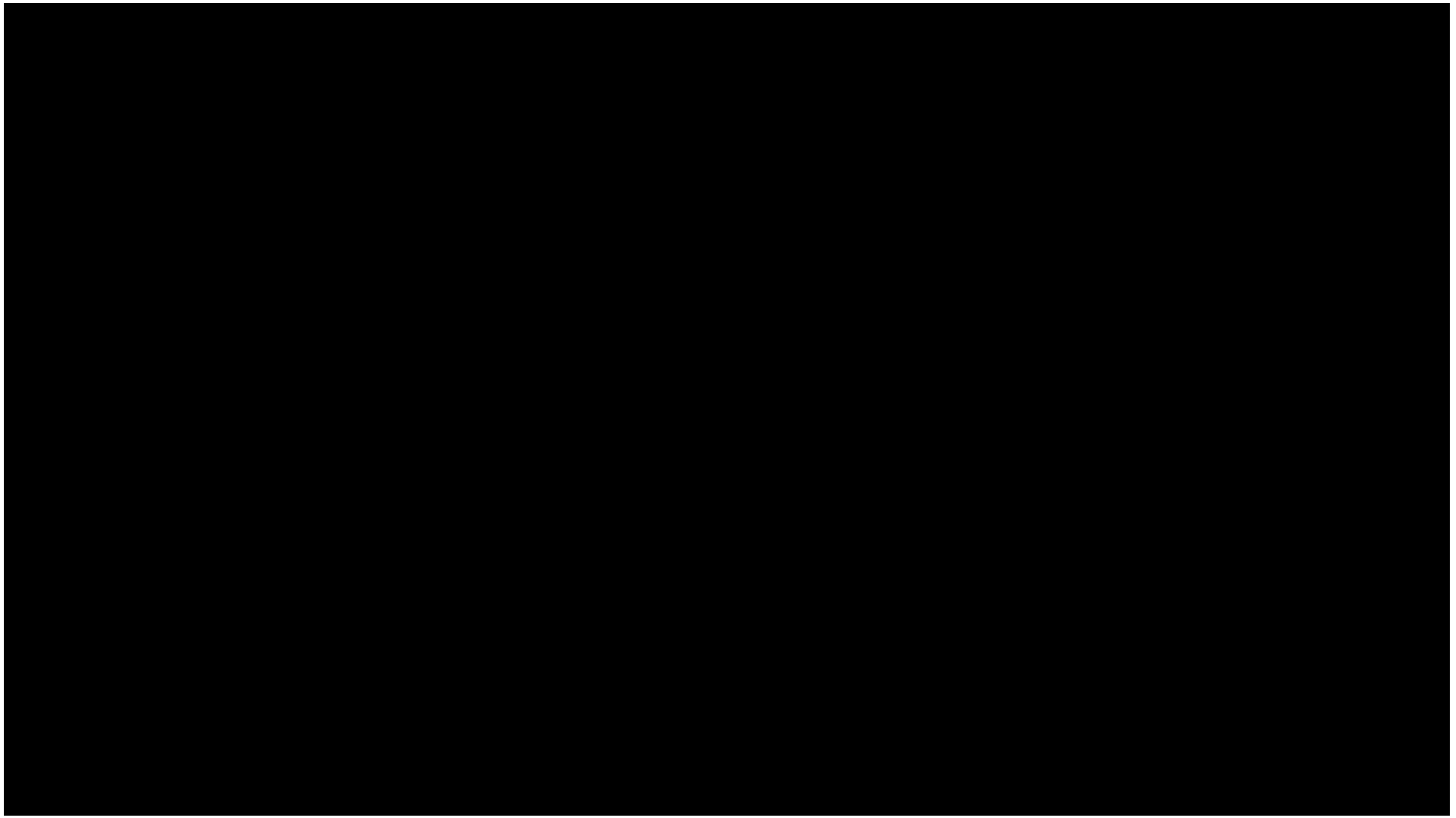
What has changed, and what has *NOT* changed, in the 70-year pursuit of Artificial Intelligence?

What is AI?

- **One of Many Definitions**
  - AI is the capability of **computational systems** to perform tasks typically associated with **human intelligence**, such as learning, reasoning, problem-solving, perception, and decision-making.
  - It is a field of research in engineering, mathematics and computer science that develops and studies **methods** and **software** that enable **machines** to perceive their environment and use learning and intelligence to take actions that maximize their chances of achieving defined goals.

Russell, Stuart J.; Norvig, Peter (2021). Artificial Intelligence: A Modern Approach (4th ed.). Hoboken: Pearson. ISBN 978-0-1346-1099-3.

What is AI?

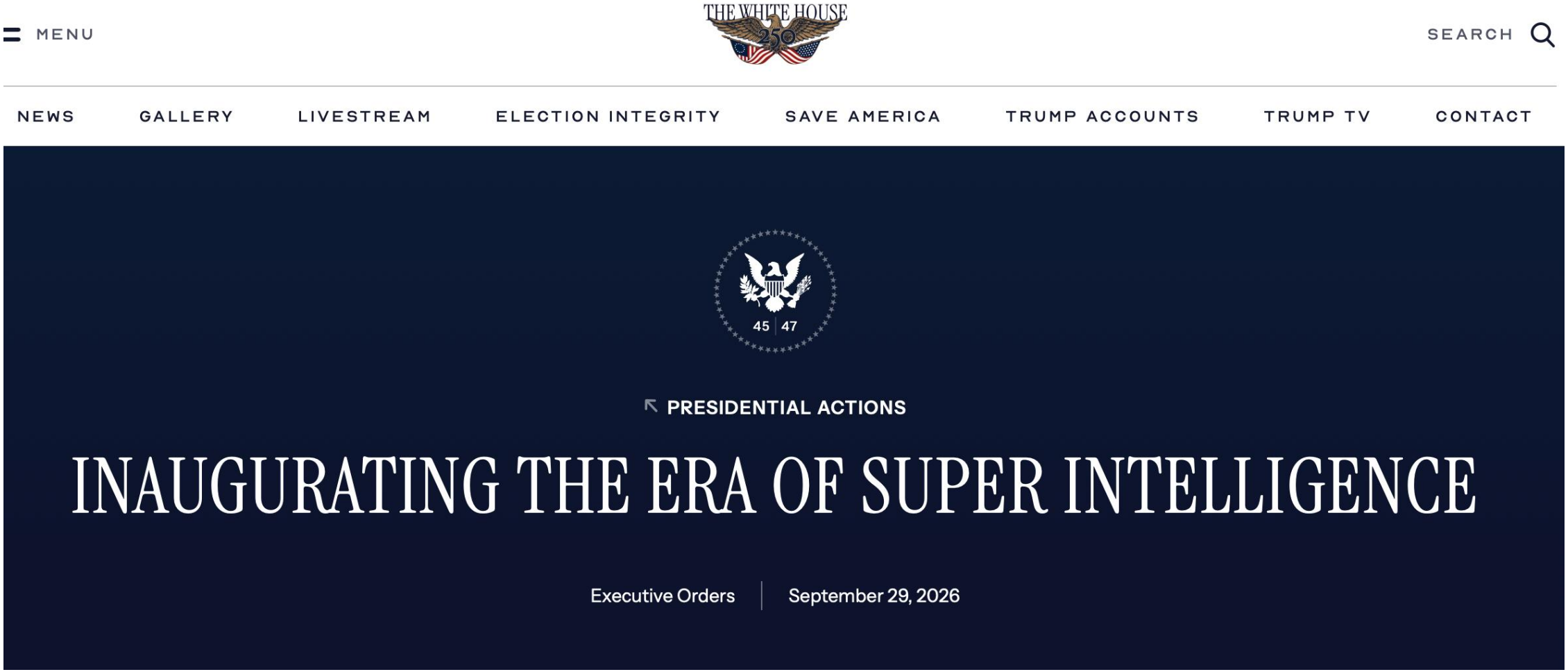


What is AI?

AI is not a single technology, but a goal-oriented field of research. At different stages in its history, researchers have understood "intelligence" in different ways.

- **Symbolic AI**: intelligence as reasoning
- **Machine Learning**: intelligence as learning from data
- **Deep Learning**: intelligence as representation learning
- **Generative AI**: intelligence as prediction, generation, and interaction

~~What is AI?~~ What is SI?



~~What is AI?~~ What is SI?



A Conceptual Map of AI Technologies

- **Task Level** --- What problem does AI solve?
  - **Classification** --- Image Classification, Text Classification
  - **Prediction** --- Time-Series Forecasting, Numerical Prediction
  - **Sequence Modeling** --- Language Modeling, Speech Recognition, Machine Translation
  - **Object Detection** --- Object Localization, Real-Time Detection
  - **Image Segmentation** --- Semantic Segmentation, Instance Segmentation
  - **Language Understanding** --- Question Answering, Information Extraction, Text Summarization
  - **Content Generation** --- Text Generation, Image Generation, Audio Generation, Video Generation
  - **Decision Making** --- Planning, Control

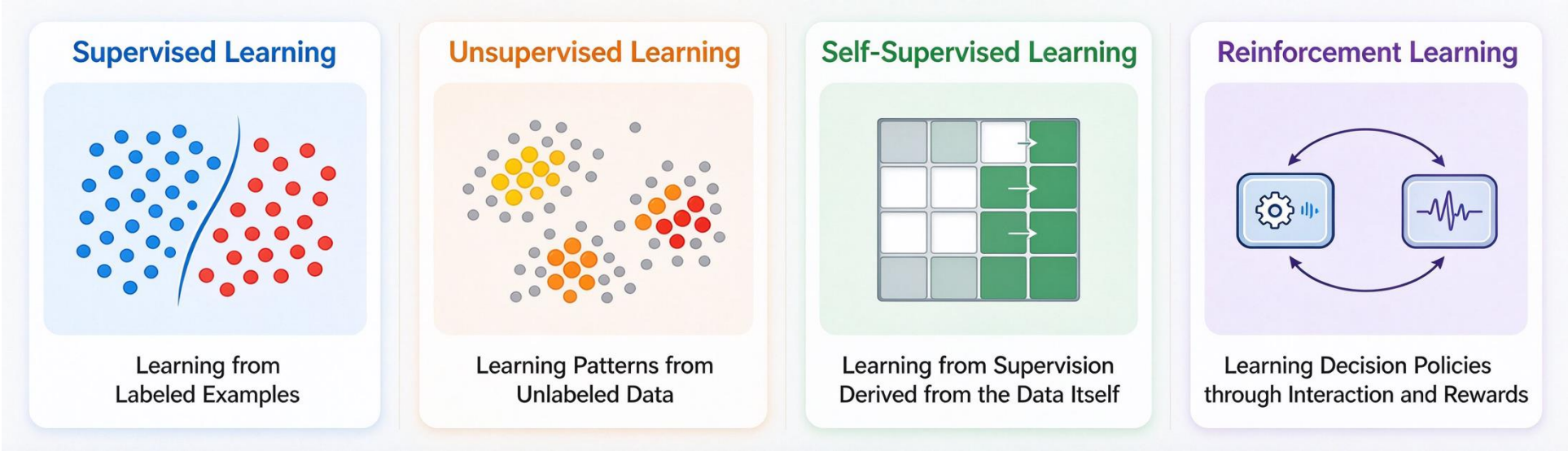
A Conceptual Map of AI Technologies



- **Architecture Level** --- How does AI process information?
  - **Multi-Layer Perceptron (MLP)** --- Feedforward Neural Network
  - **Recurrent Neural Network (RNN)** --- Vanilla RNN, LSTM, GRU, Bidirectional RNN
  - **Convolutional Neural Network (CNN)** --- LeNet, AlexNet, VGG, ResNet, U-Net, Faster R-CNN, YOLO
  - **Transformer** --- BERT, GPT, T5, Vision Transformer, Large Language Models, Multimodal Models
  - **Autoencoder** --- Denoising Autoencoder, Variational Autoencoder
  - **Generative Adversarial Network (GAN)** --- DCGAN, CycleGAN, StyleGAN
  - **Diffusion** --- Image Diffusion Models, Audio Diffusion Models, Video Diffusion Models
  - **Graph Neural Network (GNN)** --- GCN, GraphSAGE, Graph Attention Network

A Conceptual Map of AI Technologies

- **Learning Paradigm Level** — How does AI learn?
  - **Supervised Learning** — Learning from Labeled Examples
  - **Unsupervised Learning** — Learning Patterns from Unlabeled Data
  - **Self-Supervised Learning** — Learning from Supervision Derived from the Data Itself
  - **Reinforcement Learning** — Learning Decision Policies through Interaction and Rewards

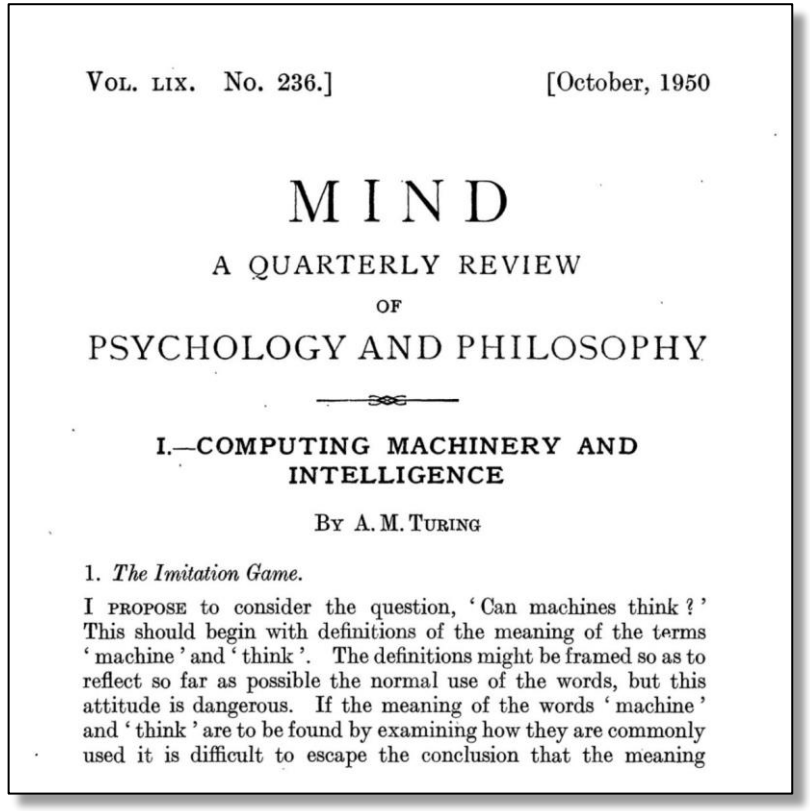


History of AI

- **Stage 1: Intellectual Foundations, 1940s–1956**
  - **Can machines think?**

    Alan Turing's 1950 paper introduced the idea of the imitation game, later commonly associated with the **Turing Test**.

    This was not yet an engineering system, but a philosophical and conceptual entry point: it made machine intelligence a legitimate question for scientific discussion.



VOL. LIX. No. 236.] [October, 1950

MIND

A QUARTERLY REVIEW

OF

PSYCHOLOGY AND PHILOSOPHY

I.—COMPUTING MACHINERY AND INTELLIGENCE

BY A. M. TURING

1. *The Imitation Game.*

I PROPOSE to consider the question, 'Can machines think?' This should begin with definitions of the meaning of the terms 'machine' and 'think'. The definitions might be framed so as to reflect so far as possible the normal use of the words, but this attitude is dangerous. If the meaning of the words 'machine' and 'think' are to be found by examining how they are commonly used it is difficult to escape the conclusion that the meaning

History of AI

THE IMITATION GAME: the real-life story of Alan Turing



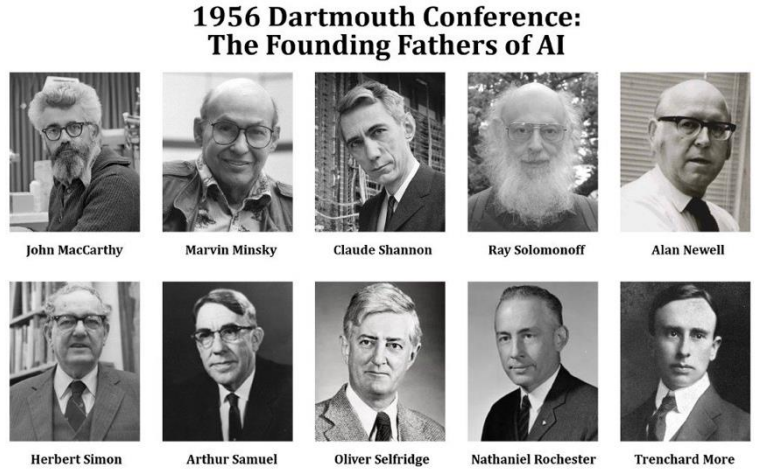
History of AI

- **Stage 1: Intellectual Foundations, 1940s–1956**
  - **Can machines think?**

    The 1956 Dartmouth Workshop --- widely regarded as the birth of AI

    John McCarthy and others used the term "**Artificial Intelligence**" and expressed a bold assumption:

    > *"learning and other aspects of intelligence could, in principle, be precisely described and simulated by machines"*

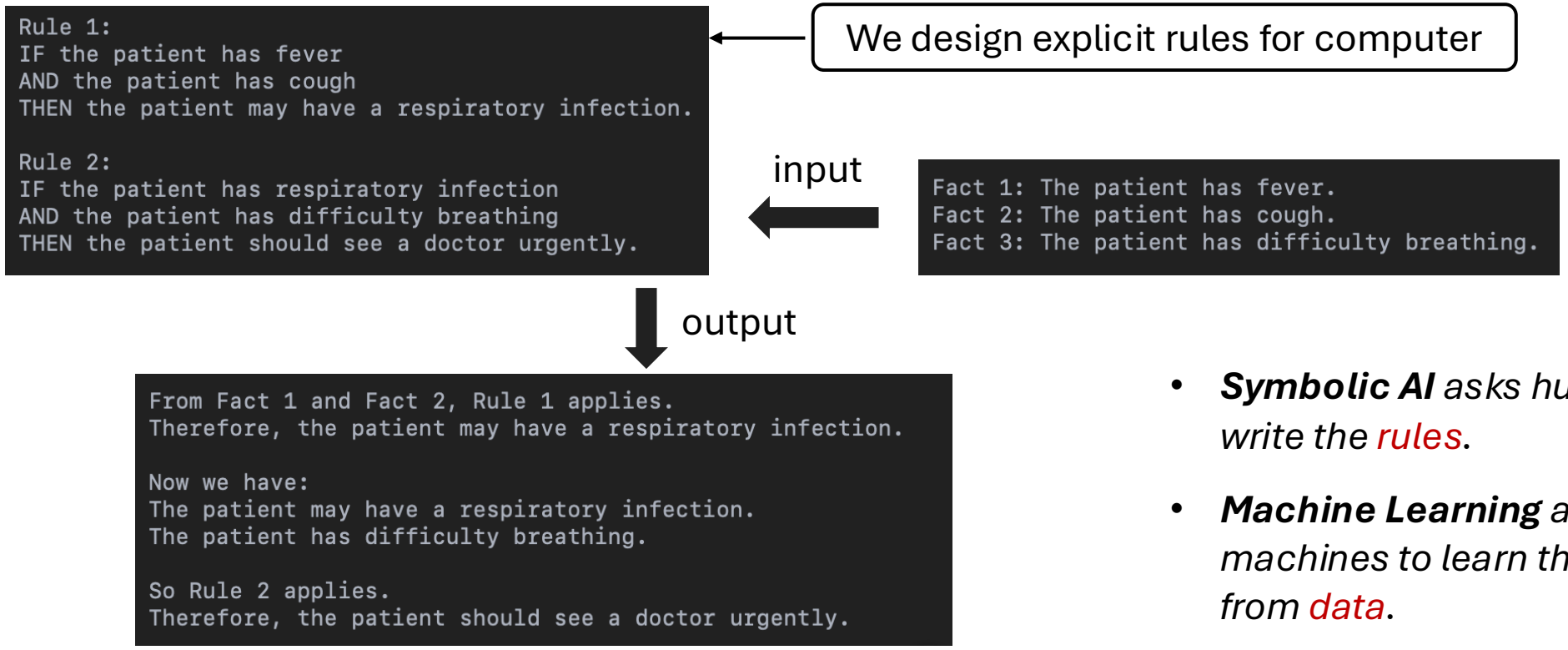




Birthplace of AI

History of AI

- **Stage 2: Golden Age of Symbolic AI, 1956–1970s**
  - **Build intelligence by explicitly representing knowledge and manipulating symbols according to logical rules.**



- ***Symbolic AI*** *asks humans to write the rules.*
- ***Machine Learning*** *asks machines to learn the patterns from data.*

History of AI

- **Stage 3: First AI Winter and Expert Systems, 1970s–1980s**
  - **First AI Winter:** Symbolic AI systems achieved impressive results in **controlled environments**, but they struggled in the **(complex and stochastic) real world**, leading to reduced funding and confidence
  - **Expert Systems** became influential, but **human expertise hard** to extract, formalize, update and encode completely.

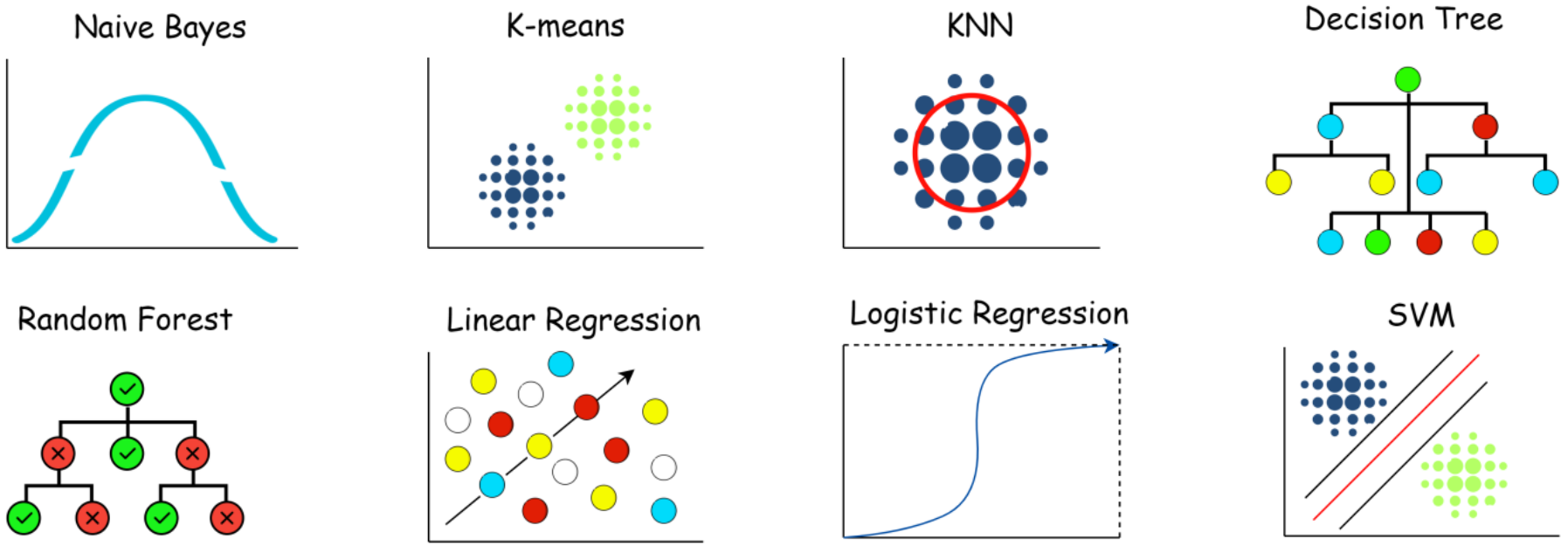
If intelligence is knowledge, can we simply write down all the knowledge?

History of AI

- **Stage 4: Rise of Statistical Machine Learning, 1990s–2000s**
  - **Instead of telling machines exactly what to do, we let machines learn patterns from data.**

    ***Examples**: decision trees, naive Bayes, support vector machines, hidden Markov models, probabilistic graphical models, and ensemble learning.*

Top Eight ML Algorithm



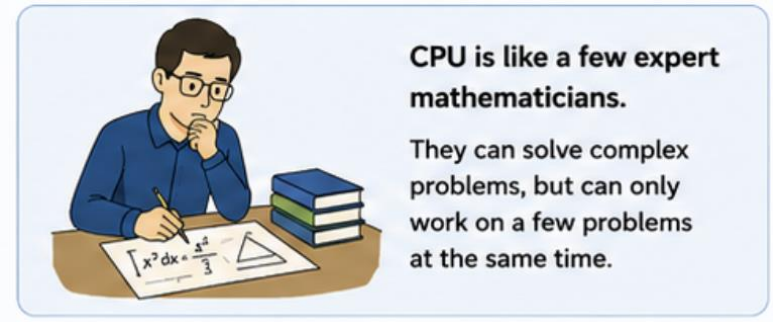
History of AI

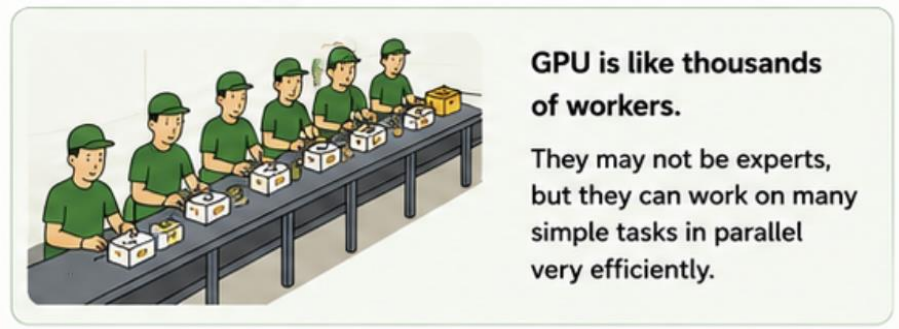
- **Stage 5: Deep Learning Revolution, 2012–2020**
  - **Graphics Processing Unit (GPUs) --- Milestone**

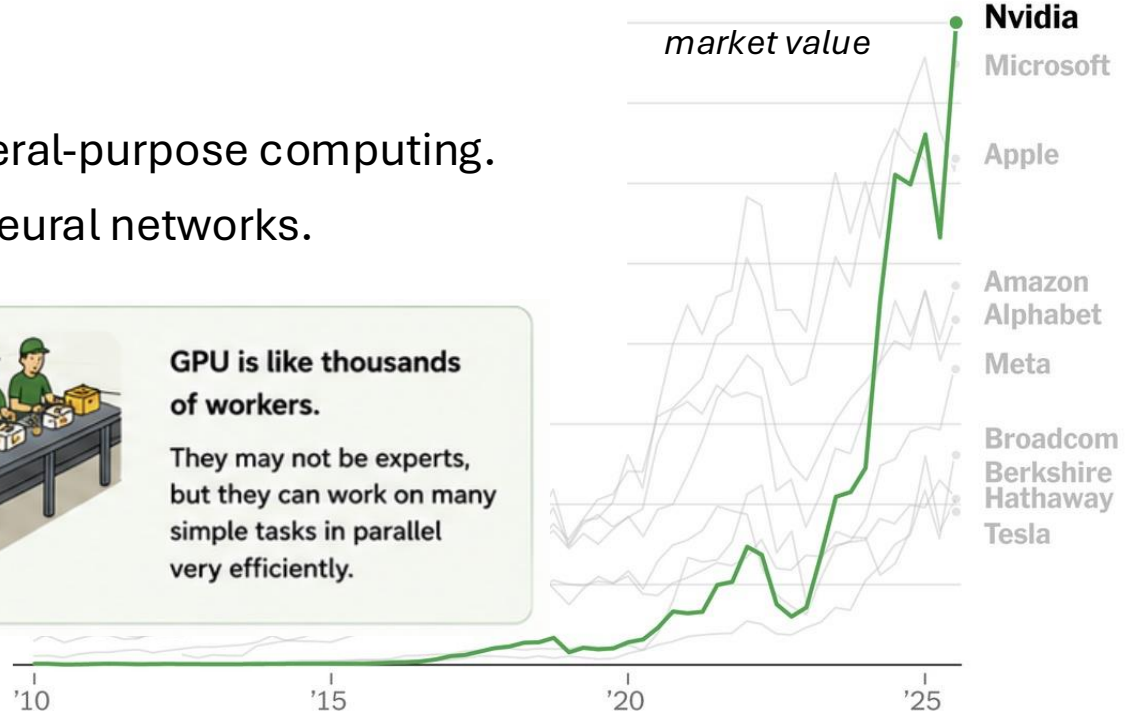
    Why GPUs Matter for Deep Learning

    - Deep learning relies heavily on matrix operations.
    - Matrix operations can be highly parallelized.
    - GPUs are designed for parallel computation.
    - NVIDIA CUDA made GPUs programmable for general-purpose computing.
    - This enabled faster training of larger and deeper neural networks.



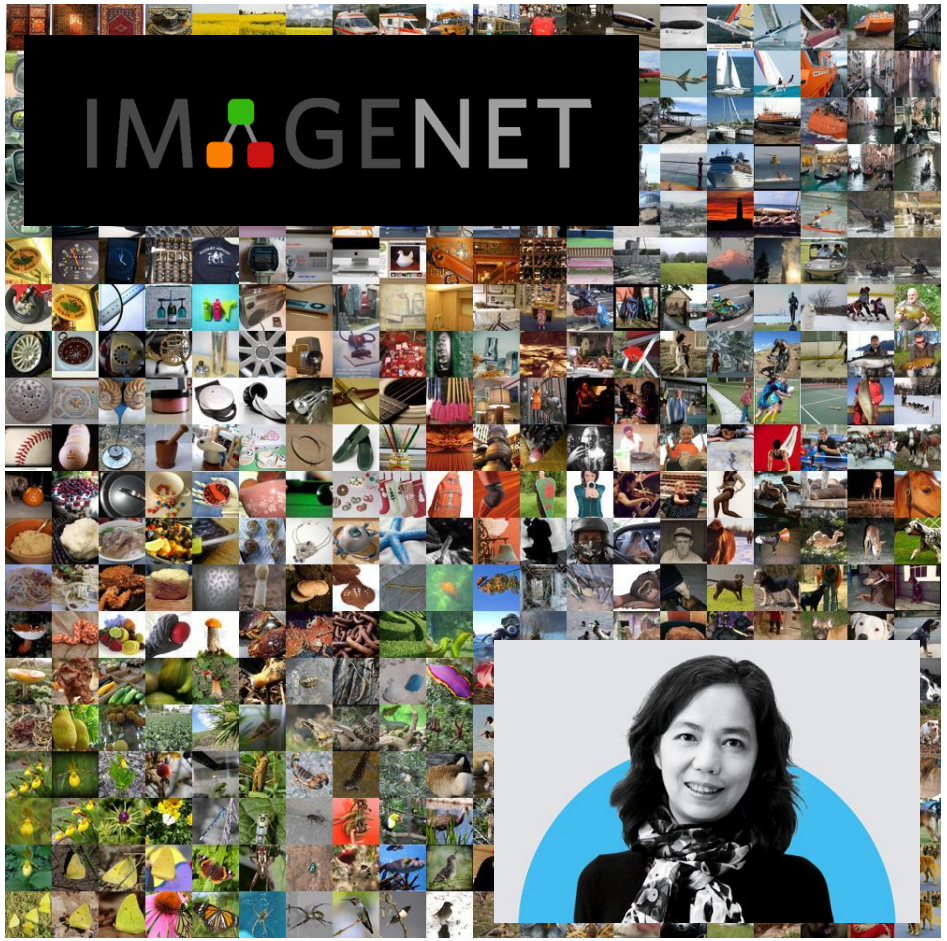






History of AI

- **Stage 5: Deep Learning Revolution, 2012–2020**
  - **ImageNet** --- **Milestone**
    A large-scale image database, containing **millions** of labeled images
  - **Significance**: It provided the data, benchmark, and competitive environment that made progress in computer vision **measurable**.
  - **Breakthrough**: In 2012, **AlexNet**, a deep convolutional neural network (CNN), dramatically outperformed traditional methods in the ImageNet challenge. This result showed the power of deep learning when combined with large-scale data and GPU computing.



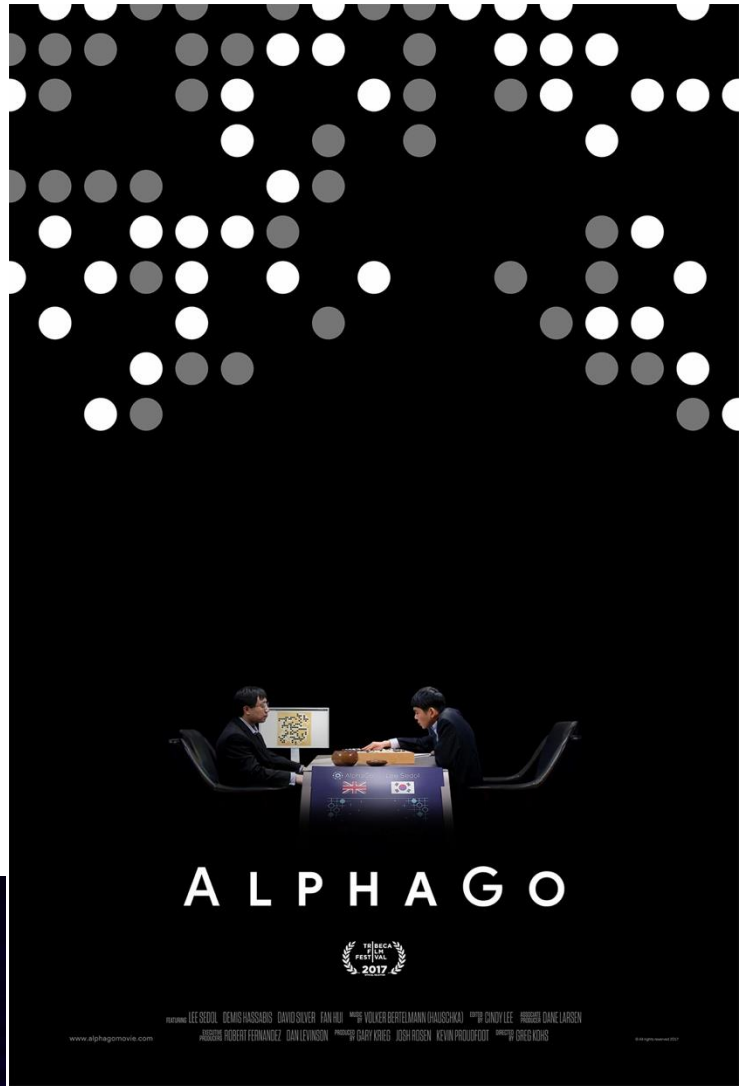
History of AI

- **Stage 5: Deep Learning Revolution, 2012–2020**
  - **AlphaGo** --- **Milestone**

    AI system developed by **Google DeepMind** to play the board game Go.

    It combines deep neural networks, reinforcement learning, and Monte Carlo Tree Search.
  - **Significance**: AlphaGo defeated Lee Sedol, one of the world's strongest Go players, in 2016. This showed that AI could master a highly complex game that was long believed to require human intuition and strategic judgment.





History of AI

- **Stage 6: Transformer, Foundation Models, and Generative AI, 2017–Present**
  - **Modern AI** (SI?) **is based on large models trained at scale and adapted to many downstream tasks**
  - **Transformer**

    A neural network architecture based on **attention mechanism**, which allows a model to process input data by focusing on the most relevant parts of the sequence.
  - **Significance**: it made large-scale language models and many modern generative AI systems **possible**



Attention Is All You Need

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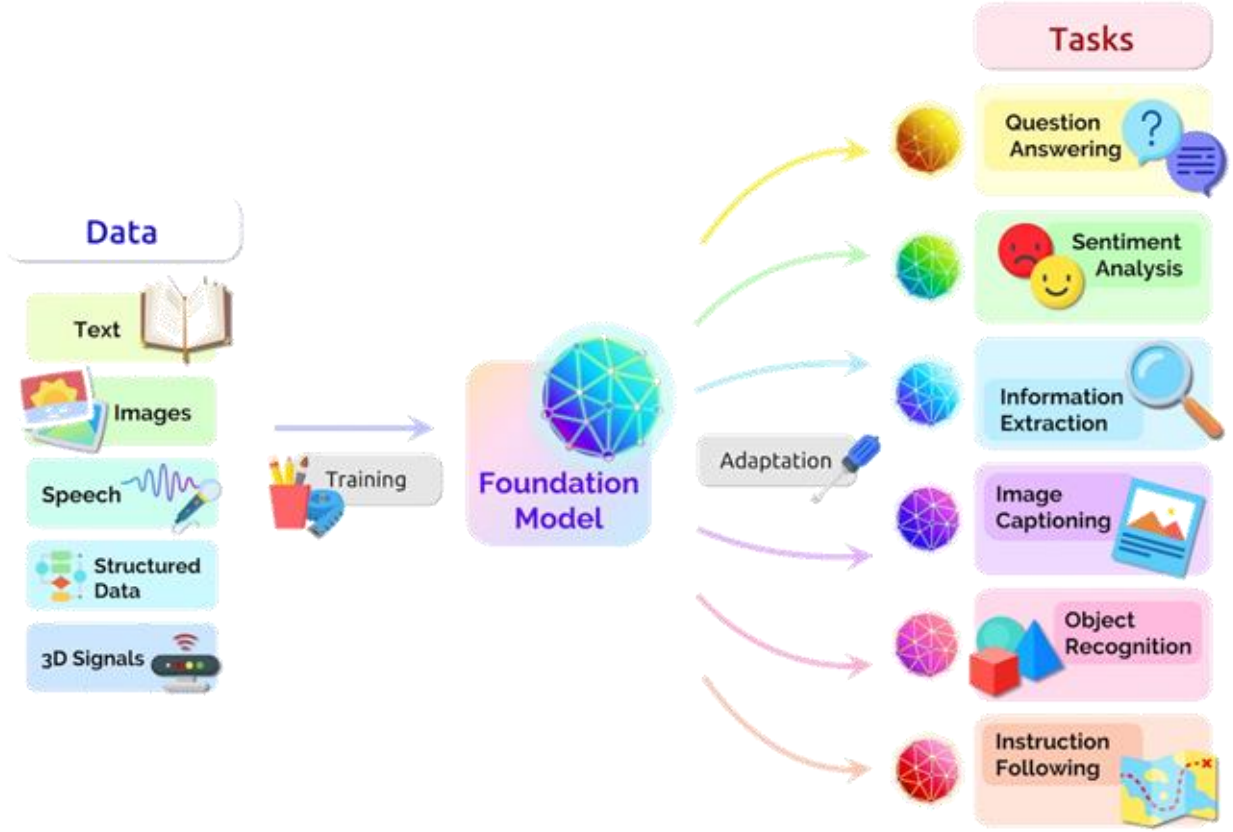
Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.8 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature. We show that the Transformer generalizes well to other tasks by applying it successfully to English constituency parsing both with large and limited training data.

History of AI

- **Stage 6: Transformer, Foundation Models, and Generative AI, 2017–Present**
  - **Foundation Models**

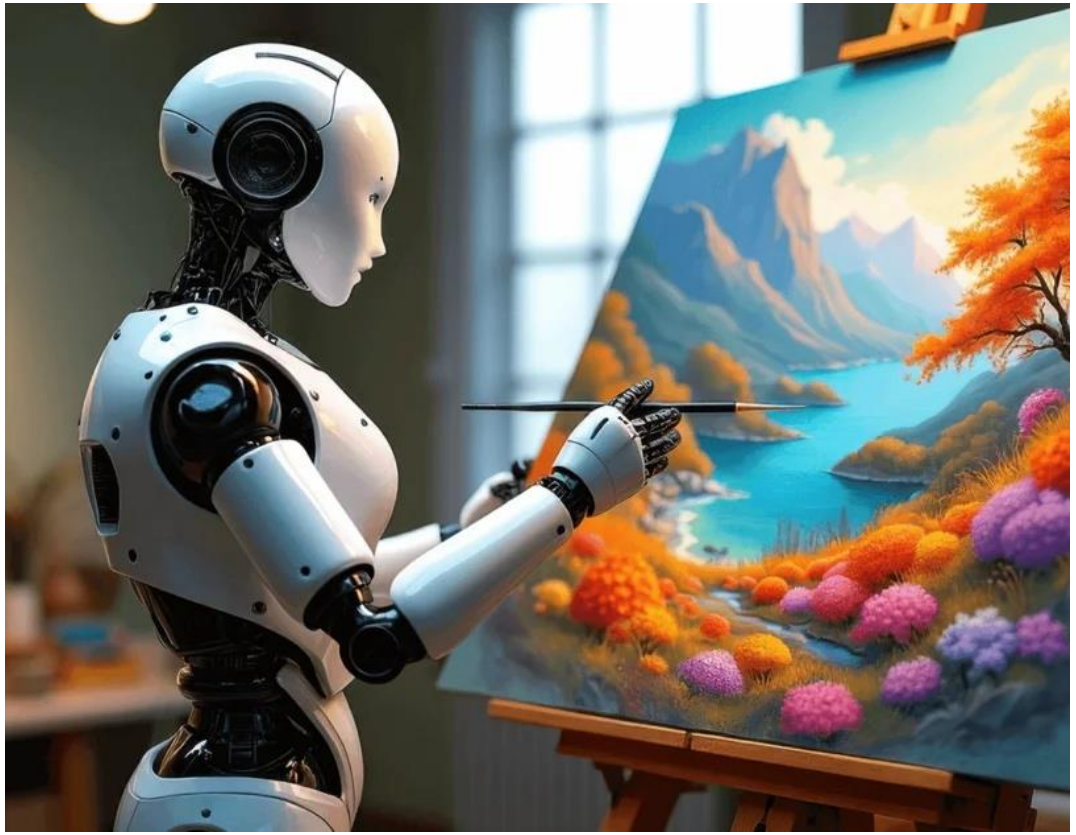
    Large AI models trained on massive amounts of broad data, which can be adapted to many different downstream tasks.
  - **Significance**: They are important because they shift AI from building separate models for specific tasks to using one general-purpose model as a reusable foundation for many applications.



History of AI

- **Stage 6: Transformer, Foundation Models, and Generative AI, 2017–Present**
  - **Generative AI**

    A type of AI that can create new content, such as text, images, code, audio, or video, based on patterns learned from data.
  - **Significance**: It changes AI from mainly recognizing or predicting information to actively producing and interacting with content in human-like ways.



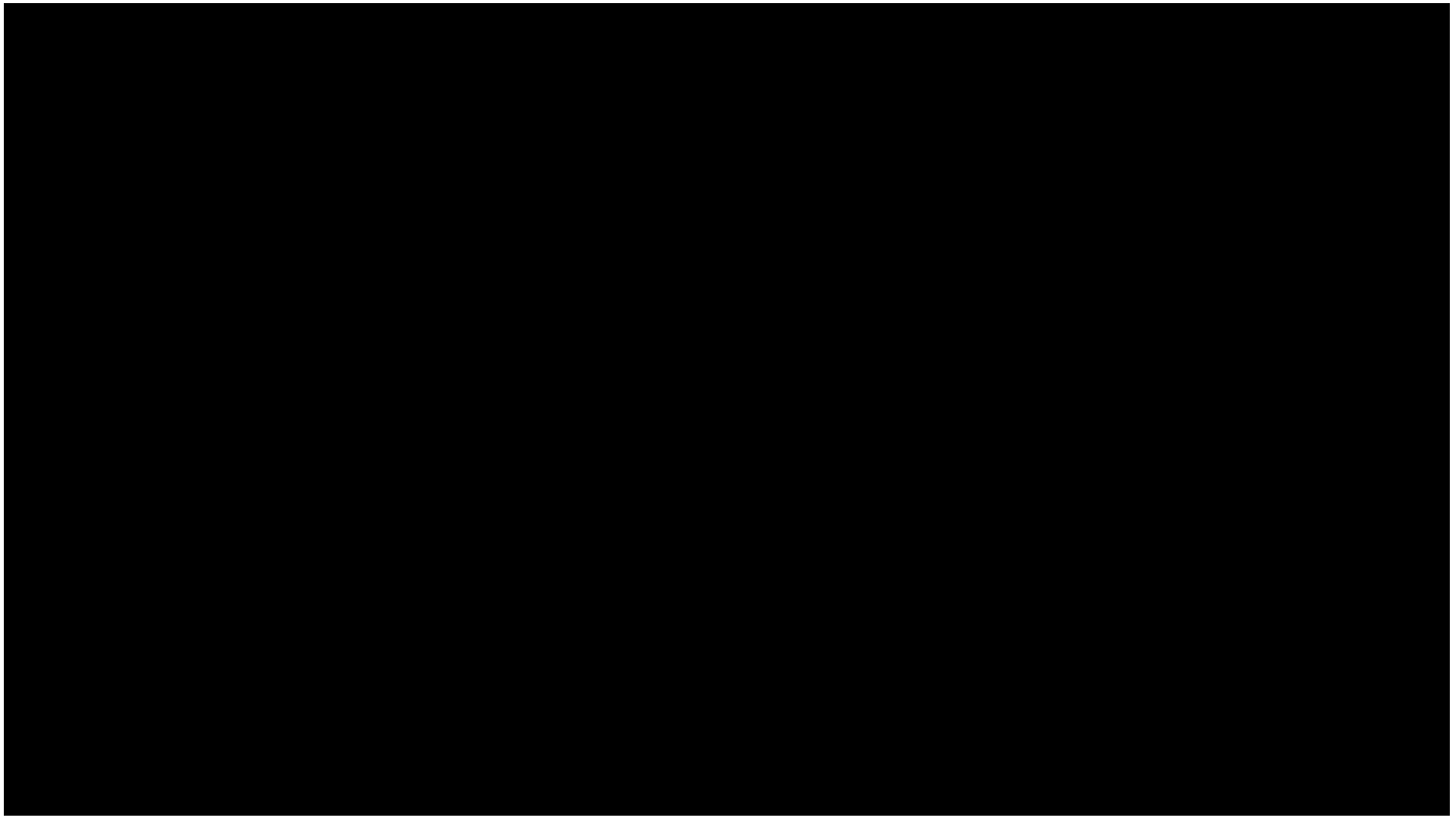
History of AI

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What is Generative AI?



History of AI

- **Stage 6: Transformer, Foundation Models, and Generative AI, 2017–Present**
  - **Generative AI**



2022 2023

In January 2025, DeepSeek released R1, showing that competitive reasoning capabilities could be achieved ***with more efficient training methods***.

Its release challenged the assumption that frontier AI must always depend on ever-larger computational budgets, triggering a sharp market reaction, including nVIDIA's historic one-day loss in market value.

History of AI



scientists

?

entrepreneurs



“0-1” ORIGINAL INNOVATION

creates a new path from scratch. It is uncertain, time-consuming, resource-intensive, and often full of trial and error

“1-100” APPLICATION INNOVATION

builds on an already validated path. It is usually more stable, faster, and more scalable, especially when supported by a large market.



Large Market Advantage

More users, greater demand, faster iteration, achieve leapfrog and surpass



China

Leverage the proven path, accelerate fast

Trial and error

Trial and error



United States

Explore first, path validated



Smaller Market

Fewer users, lower demand, slower adoption, harder to surpass

Starting Point

U.S.: Explore first, optimize later

Constant trial and error to find the right path

China: Leverage the proven path, accelerate fast

Build on validated path and surpass

History of AI: Summary

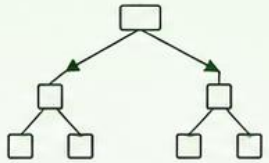
1 Intellectual Foundations
1940s–1956





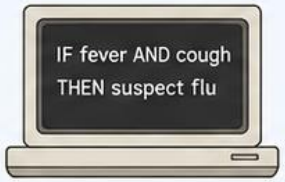
- Can machines think?
- Turing Test (1950)
- Early ideas on machine intelligence
- 1956 Dartmouth Workshop: "Artificial Intelligence" is coined

2 The Golden Age of Symbolic AI
1956–1970s



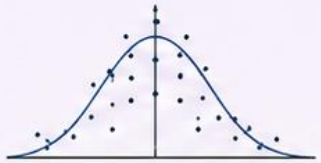
- Intelligence as reasoning with symbols
- Search, logic, planning
- Theorem proving
- Game playing
- Knowledge representation

3 The First AI Winter and Expert Systems
1970s–1980s



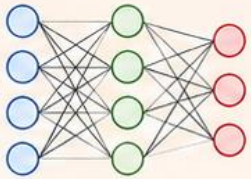
- Limits of symbolic AI
- Overpromises, fragile systems
- Reduced funding (AI Winter)
- Rise of expert systems
- Knowledge acquisition bottleneck

4 The Rise of Statistical Machine Learning
1990s–2000s



- From programming to learning
- Probabilistic models
- Support Vector Machines
- Ensemble methods
- Data-driven approach

5 The Deep Learning Revolution
2012–2020

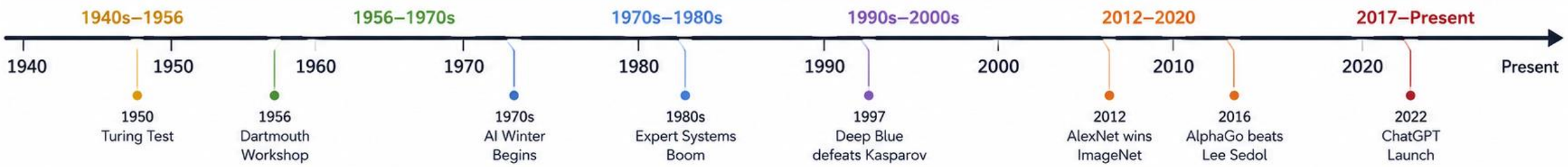


- Data + GPUs + neural networks
- Representation learning
- Breakthroughs in vision, speech, language, and more
- AlphaGo (2016)
- AlphaFold (2020)

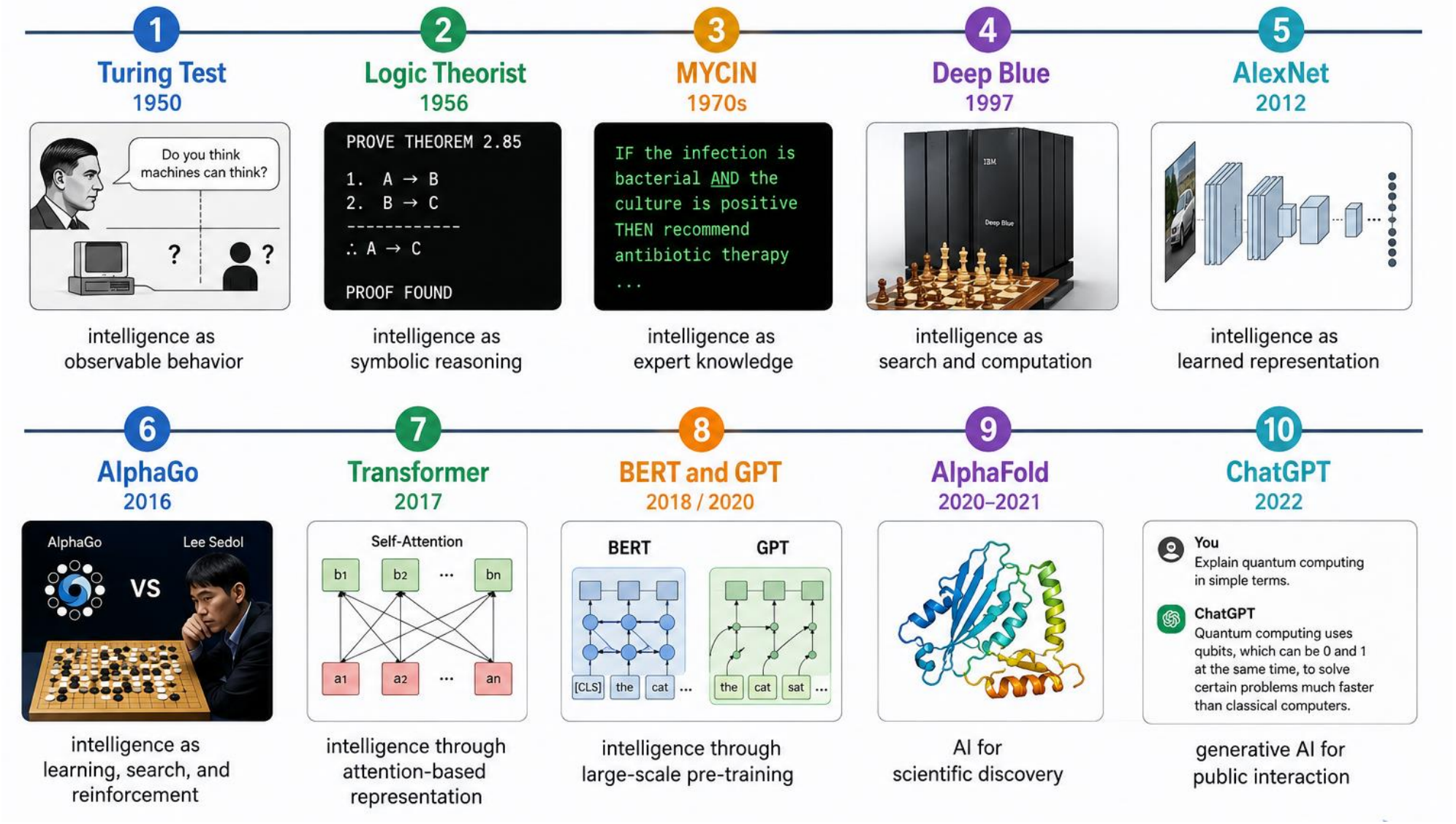
6 Transformers, Foundation Models & Generative AI
2017–Present



- Transformer architecture (2017)
- Foundation models at scale
- Generative AI: text, code, image, audio, video
- ChatGPT (2022) and beyond
- New opportunities and challenges



History of AI: Summary



What has changed, and what has *NOT* changed, in the 70-year pursuit of Artificial Intelligence?

History of AI: Summary

AI in the Past 70 Years: What Has Changed, and What Has Not Changed?

From the 1950s to Today

WHAT HAS CHANGED

1. METHODS
From symbolic reasoning and hand-written rules to statistical learning, deep neural networks, foundation models, and generative AI.

Rules → Learning

2. RESOURCES
From small datasets and limited computing to massive data, large-scale computing power, advanced algorithms, and global systems.

Small → Massive

3. APPLICATIONS
From laboratory demonstrations and narrow expert systems to search engines, recommendation systems, autonomous systems, scientific discovery, and everyday human-AI interaction.

Lab → Real World

4. SCALE AND IMPACT
From a research topic in academia to an industrial platform, economic force, and social infrastructure that touches billions of lives.

Research → Societal Impact

WHAT HAS NOT CHANGED

1. CENTRAL AMBITION
The fundamental goal remains the same: to build machines that can perform tasks associated with human intelligence.

2. CORE QUESTIONS
Can machines reason? Can they learn?
Can they understand language?
Can they act reliably in the real world?

3. LIMITATIONS REMAIN
AI systems still struggle with common sense, robustness, explanation, trustworthiness, and alignment with human values.

4. THE FUNDAMENTAL TENSION
AI is always a balance between scientific discovery and engineering application, between 0-to-1 innovation and 1-to-100 scaling.

FINAL TAKEAWAY

Over the past 70 years, AI has changed dramatically in methods, scale, and applications, but its deepest goal remains the same: **understanding** and **building intelligence**.

Thank You

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